

Animesh Tripathi

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Location: Bengaluru, India

DATA SCIENTIST

As a passionate and results-driven **Data Scientist**, I bring a unique blend of analytical skills, machine learning expertise, and a proven track record of transforming complex datasets into actionable insights. With **3 years** of hands-on experience with Deep Learning, NLP, LLMs, and good knowledge of ML tools and frameworks. I am adept at leveraging advanced statistical models and machine learning algorithms to solve real-world problems and drive business success.

EDUCATION

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY VADODARA

M.TECH in Computer Science & Engineering with specialization in DATA SCIENCE

Gujarat, India

Aug 2022 – Jul 2024

JSS ACADEMY OF TECHNICAL EDUCATION NOIDA

B.TECH in Computer Science & Engineering

Noida, India

Aug 2018 – Jul 2022

EXPERIENCE

SENIOR DATA SCIENTIST

ENVERUS PRODUCT AND INNOVATION, Bengaluru, India

Dec 2025 – Current

Domain-Specific Small Language Model for Instant Analyst

Designed and fine-tuned a custom Small Language Model (SLM) to replace prompt-engineered nodes in an agentic Instant Analyst architecture, reducing inference latency and improving task-specific accuracy through supervised fine-tuning and architecture optimization.

ASSOCIATE DATA SCIENTIST

PHILIPS INNOVATION CAMPUS, Bengaluru, India

Jun 2023 – Dec 2025

Oral Healthcare Chatbot - MultiAgent Multimodal RAG Based AI Tool

Developed a LangGraph-powered, multimodal, multi-agent AI system to extract actionable insights from Oral Healthcare's extensive project archives (2015–present). Enabled intelligent, query-driven retrieval across diverse file formats—CSV, Excel, DOCX, PPTX, and PDF—by orchestrating information from multiple vector databases, significantly improving knowledge reuse and decision-making efficiency.

TextSense AI – Agentic AI Assistant for Redshift Database Insights

Developed TextSense AI, an intelligent assistant powered by a LangGraph-based SQL agent capable of autonomously generating and executing SQL queries to extract insights from databases. The system interprets query results and presents them in both natural language and structured tabular formats, significantly enhancing data accessibility for non-technical users. By automating SQL query generation, execution, and interpretation, TextSense AI streamlines data exploration and accelerates data-driven decision-making.

Data Engineering in Personal Health

Strategically Orchestrated ETL Processes

Led the strategic design and execution of multiple Extract, Transform, Load (ETL) processes, ensuring a seamless and efficient flow of data using Apache Airflow.

Implementation of new system for GDPR Compliance and Service Security Risk Assessment

Independently implemented data retention pipelines for General Data Protection Regulation (GDPR) compliance. Additionally, designed and deployed robust data attestation pipelines ensuring data security and regulatory adherence within the organization under Service Security Risk Assessment(SSRA).

Architected, Tested, and Deployed Robust Data Pipelines:

Led the development life cycle, designing, testing, and deploying robust data pipeline architectures. Demonstrated solutions that seamlessly handle data flow from source to destination. Collaborated with senior management to gain a comprehensive understanding of company objectives.

PROJECTS

Generative AI powered SQL Agent for Structured Database

[Source Code](#)

- * Developed a conversational SQL agent using LangGraph to enable natural-language querying over structured databases.
- * Implemented an agentic workflow with a LangGraph state graph, including nodes for table listing, schema retrieval, query generation, error-checking, and execution.
- * Separated natural-language reasoning from SQL generation to ensure more accurate and context-aware response.
- * Designed a robust error-checking phase to detect and fix common SQL issues such as NULL handling, quoting, and joins.
- * Integrated a Streamlit-based UI for interactive conversations, enabling real-time query execution and streaming of results to users.

Generative AI powered Counselling Assistant

[Source Code](#)

- * Created a Assistant using Retrieval Augmented Generation to help student in there counselling process in National Institutes.
- * Generated a knowledge base of documents using Facebook AI Similarity Search (FAISS) .
- * Added a Robust Guardrail system to restrict model from hallucination.
- * Optimize the inference time of LLM using vLLM.
- * Developed a system that responds to questions based on the context retrieved from the knowledge base.
- * Created a user-friendly web application using Streamlit framework.

Extraction of River-Network from Satellite Image

[Source Code](#)

- * Created a deep learning model that can identify River networks from satellite images.
- * Performed Data-Augmentation technique to enhance CNN models' generalization by increasing training data diversity.
- * Evaluated model's performance on various metrics.

TECHNICAL SKILLS

Languages : Python, C, SQL

Frameworks : LangGraph, LangChain, Apache Airflow, Streamlit

Libraries : Tensorflow, Pytorch, Transformer, Sklearn, Pandas, Pyscopg2, Boto3, Pyspark, Pytest

Databases : Amazon Redshift, PostgreSQL

Dev Tools : Visual Studio Code, Git, Github, Azure DevOps, CI/CD

ACHIEVEMENTS

- GATE 2022 Qualified
- B.TECH first-year College Topper
- Published a Research Article on Extraction of River Network from Satellite Image

CERTIFICATION

- [Microsoft GenAI Hackathon](#)
- [NLP with Python for Machine Learning Essential Training](#)
- [Advanced Learning Algorithms](#)
- [Supervised Machine Learning: Regression and Classification](#)